Do brains synchronize more to biological motion? An inter-subject correlation (ISC) analysis of EEG whole-body movement data

Introduction

People are highly sensitive to biological motion, the dynamic visual patterns associated with the movements of living entities. For example, we can recognize human movement from just a few moving dots placed on the major joints of the body (Johansson, 1973), known as point-light displays (PLDs). At the neural level, the response to biological motion has been well characterized, implicating visual and sensorimotor regions (Grosbras et al., 2012). Research using electroencephalography (EEG) and magnetoencephalography (MEG) has found brain responses to biological motion from roughly 200-360 ms after stimulus onset (Chang et al., 2021; Hirai et al., 2003; Jokisch et al., 2005). In such studies, brain activity is typically measured at the individual level for biological motion versus a control condition. For example, "scrambled" PLDs with the dot starting locations randomized provide a control for low-level visual motion while removing the sense of global biological motion. For each individual, responses are averaged within each channel and time window, and these averages are then compared between conditions using inferential statistics (e.g., *t* tests, ANOVA).

However, averaging the brain response across a stimulus shows how strongly the brain responds, but not whether the timeline of that response is the same across individuals. Research using naturalistic stimuli, such as audiovisual movie clips, suggests that dynamic, socially relevant stimuli evoke neural responses with a shared temporal structure across viewers (Nastase et al., 2019). In fMRI studies, the shared pattern of neural activity across individuals has been linked to the effectiveness of interpersonal communication (Stephens et al., 2010). More recently, this shared response has also been measured with EEG, where it

can provide information about attentional engagement (Dmochowski et al., 2012) and value judgments (Dmochowski et al., 2014).

Intersubject correlation (ISC) measures how similar the activity in a given brain region is across people over time while they experience the same naturalistic stimulus (Hasson et al., 2004, 2010). Participants receive the same time-locked stimulus, which allows the degree of shared response to be compared between experimental conditions. To do this, the response time course of each brain region or electrode in one subject is correlated with the corresponding time courses of the other subjects, rather than correlating time series across different electrodes within a single subject (Nastase et al., 2019; Simony et al., 2016). This makes ISC analyses less susceptible to idiosyncratic physiological noise and head motion than conventional functional connectivity analyses. However, ISC cannot detect idiosyncratic responses, and a low ISC means that a response is not consistent across people, rather than that the brain is not responding.

Biological motion can be analyzed using ISC as it carries dynamic social information. However, it remains unclear whether PLDs, which reduce biological motion to a few moving dots, elicit the same kind of shared neural response as more complex naturalistic stimuli.

This project will try to address this question by measuring ISC in EEG data recorded while participants viewed PLDs of coherent, whole-body actions and “scrambled” versions of the same stimuli in which the starting locations of the dots were randomized. Scrambling keeps each dot’s movement the same but breaks up the shape of the body, so comparing the two conditions shows the effect of seeing a whole human figure. ISC, which quantifies how closely each participant’s neural activity tracks that of the other viewers, will be computed in MATLAB using Dmochowski’s Reliable Components Analysis (RCA) toolbox (Dmochowski et al., 2012; Dmochowski, Greaves, & Norcia, 2015) and compared between the two conditions. EEG electrodes record a mixture of many neural sources, some of which

are noise, and RCA combines the electrodes into a few components that make the response as consistent as possible across participants. This concentrates the stimulus-driven signal, reduces the number of statistical comparisons, and gives each component a scalp topography showing where the shared response is strongest.

We hypothesize that ISC will be higher for coherent than for scrambled motion. Higher ISC would not mean that coherent motion simply produces more brain activity, but that more of the neural response is driven by the stimulus in the same way across viewers, suggesting that the human form itself is processed consistently across people.

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